

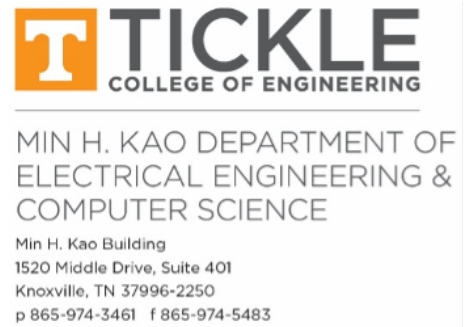
SYLLABUS: COSC 594 – FALL 2025

Course Title: Interplay Between RL and GenAI

Lecture Time: Mon, Wed, Fri, 11:30 am - 12:20 pm

Lecture Location: Min Kao Building 406

Instructor: Dongsheng Ding (Email: dongshed@utk.edu)



Course Description: This course explores foundational research at the intersection of reinforcement learning (RL) and generative artificial intelligence (GenAI). This course has three main objectives: (1) to introduce reinforcement learning for generative models such as language models (LMs) and diffusion models (DMs); (2) to examine the use of generative models in reinforcement learning; (3) to identify key limitations, open challenges, and promising research directions at this intersection. The course emphasizes a balanced understanding of theoretical aspects and practical algorithms.

Prerequisites: A foundation of linear algebra, probability, calculus, and convex analysis is required. Familiarity with machine learning and/or reinforcement learning is recommended.

Required Materials: There is no required textbook. All course material will be shared in class and/or provided online.

Office Hours: After class, or by appointment.

Tentative Course Outline:

- Overview of RL and GenAI (1 week)
- Policy optimization (1 week)
- Variational inference (1 week)
- Alignment and composition of LMs (2 weeks)
- Alignment and composition of DMs (2 weeks)
- Unlearning for GenAI (1 week)
- DMs for (policy) optimization (1 week)
- DMs for robotics (1 week)
- Other topics (1 week)
- Student presentation (1 week)
- Report writing and feedback (1 week)

Grading: 10% Class participation, 40% Homework (4 assignments, each 10%), 50% Project (20% presentation and 30% report).

- Class participation includes asking questions, engaging in discussions, scribing lectures, and providing feedback.
- There are four homework assignments that are equally weighted. Discussions with others are allowed, but all answers must be completed independently. Any use of ChatGPT must be explicitly acknowledged, and you are responsible for the correctness of your answers. Homework is due two weeks after it is assigned.
- Project aims to present and discuss a machine learning paper on the interplay between RL and GenAI. Students will work in pairs to present the paper and summarize it in a joint report in $\text{\LaTeX}$. The paper and presentation date will be assigned. The final report is due the last day of class.

Honor Statement: <https://studentlife.utk.edu/studentconduct/academic-dishonesty/>